# The Famous London Cholera Map (1854)




Many people died in Europe during the deadly and recurring cholera epidemics. In times before modern bacteriology, medicine, epidemiology and public health, there was great debate and misunderstanding about the causes and spread of this disease.

In this exercise, we will re-create Dr. John Snow's famous 1854 map of the Soho region of London, one of the hardest hit spots during this cholera epidemic. At the time of this epidemic (one of several in Europe and London) during the 1840's and 1850's, Dr. Snow had already become well known through his pioneering work in use of chloroform (having helped deliver two of Queen Victoria's children) and his early work in what has become the field of epidemiology.

At the time, it was generally assumed, and accepted, that cholera was an airborne disease (miasma). Given the open sewers in London (and elsewhere), the presence of livestock, the crowded conditions, and the general lack of proper sanitation, there was

plenty of reason to believe the miasma theory. Dr. John Snow, a resident of the area, thought otherwise.

Instructions:

Sign into your ArcGIS Online Account:
https://carnegiemellon.maps.arcgis.com/

Create a New Map (select Map from the top row of menu items).

Under Layers / Add / select 'ArcGIS Online' and enter *cholera* in the search box. Many results will be returned. To narrow your search, you can use the 'owner' keyword. Add the clause *owner:sappleton_Education* to your search criteria. Three layers should be returned: Snow's Cholera Map, Cholera Cases and Water Pumps, all authored by Sarah Appleton.

Add the layers: "*Cholera Cases*", "*Water Pumps*", *and* "*Snow's Cholera Map*" *l*ayers. (The '<' returns you to the layers pane). Reposition layers, if necessary, so that the Water Pumps layer is on the top and Snow's Cholera Map is on the bottom.

*Snow's Cholera Map* layer shows the Soho neighborhood as it would have mapped in 1854. It is a digital, or 'raster' image, and has been 'georectified' to match the base map's spatial grid. He used tick marks in his original map to indicate the number of deaths at each location. *Cholera Cases* shows the addresses where cholera deaths occurred along with the number recorded. *Water Pumps* shows the location of water pumps throughout the area. Click on any point to see relevant data.

Zoom to the extent of John Snow's Map and create a bookmark (sixth option on the left hand command bar) called 'Soho 1854'. This will make your map easier to find in the future. Save your map with a descriptive name (disk icon on the left with blue dot.).

Experiment turning your layers on and off individually. Clicking on '…' in the Water Pumps or the Cholera Cases layer brings up some additional options. Show the table for each to see the underlying data.

Next, we will do some styling to better visualize the situation. Select Layers and highlight the Cholera Cases layer. It should be highlighted in gray. We will style the cholera deaths so that the symbols are in proportion to the number of deaths at each location. Select the styles option on the right hand command bar. Under 'Choose attributes' add the field 'Deaths' as the attribute to show. The style changes to 'Counts and Amounts' as circles. Click on 'Style options' and style symbols so that the maximum size is 30 px. Experiment with different colors or sizes so the map is easy to read and interpret. 'Done' when ready to go on. Save your map with a descriptive name and, optionally, with useful tags.

Open the table for Cholera Cases ( ... Show table) and carefully examine the data. Note the attributes. Sort, in descending order, by number of cases. What is the top address

and how many deaths were there at this address? Select the record (check the box) and note it's location on the map. You can zoom to this location with the '… Zoom to selection' option (header row of the table). Zoom back out (you can use your bookmark here!). Locate the next two most deadly addresses and zoom in. Note that the rows you select will be faintly highlighted on your map.

Finally, style the Cholera Cases layer with a heat map and see if you can locate the epicenter of this outbreak of cholera. Adjust the Area of influence so that your map is most illustrative. Save your map.

What might you (or Dr. Snow) conclude? How would you interpret Dr. Snow's original map? Can you explain why certain parts of the areas in question seemed less likely to be infected?

Experiment with controls or style options as you wish and save your map frequently when you reach a good checkpoint. (There is no 'easy' undo control.)